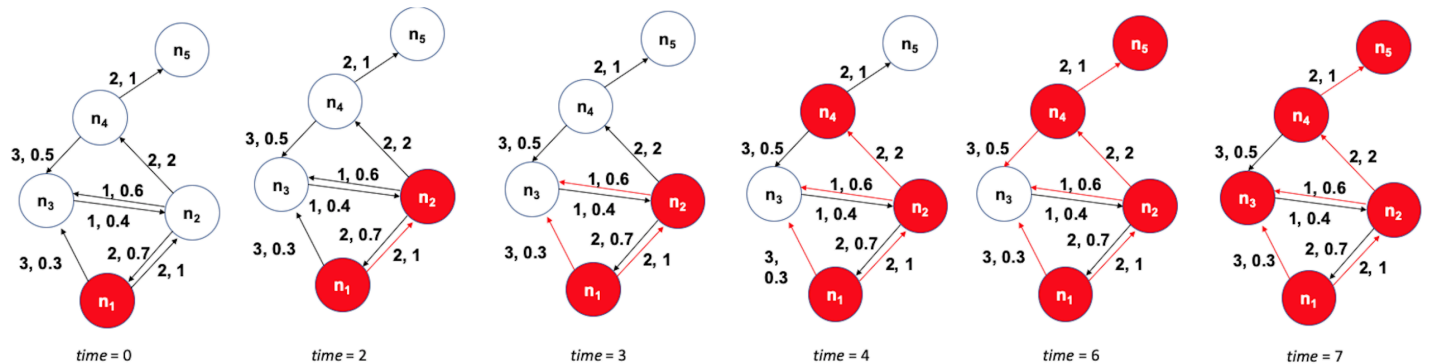


L10: Asynchronous Linear Threshold (ALT) Model

The unweighted Linear Threshold model assumes that a “**node**” (*brain region in this case*) becomes active when more than a fraction of the neighboring nodes it receives incoming connections from being active.



Here, we use a variation of this model with a) weighted connections, where the weights are based on the connection density of the projections, and b) connection delays, where the delays are based on the physical distance between connected brain regions.

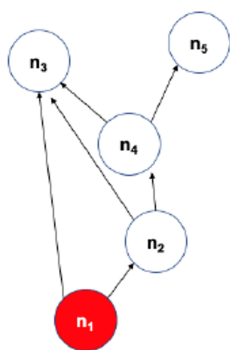
The state of a node- i is initially $s_i(t) = 0$. It becomes equal to 1 when:

$$\sum_{j \in N_{in}(i)} w_{ji} s_j(t - t_{ji}) > \theta$$

where $N_{in}(i)$ is the set of nodes that node- i receives incoming connections from. w_{ji} and t_{ji} are the weight and delay of the connection from node- j to node- i , respectively, and θ is the activation threshold.

The ALT model is simple yet it incorporates information about distances between brain regions (to model connection delays) and uses local information (*a thresholding nonlinearity*) to potentially gate the flow of information.

The previous visualization shows a toy example with five nodes. The source of the cascade is node n_1 . Each edge shows two numbers, the first is its delay, and the second is its weight. The activation threshold is set to $\theta = 1$. The cascade takes 7-time units to propagate throughout the whole network.



The visualization at the left shows the DAG that represents the activation cascade. For example, the activation of node n_3 takes place only after nodes n_1 , n_2 , and n_4 become active.